What gets shown — and what gets overlooked?

Lionael Surya and Margil Parekh

We matched **8.6 million** logged article impressions from one week of US Microsoft News (MSN) against the **65,238** articles in the dataset, and the two don't line up. **Entertainment** got 5.1× its share of the catalogue; **sports**, 30% of all articles, got 0.37×.

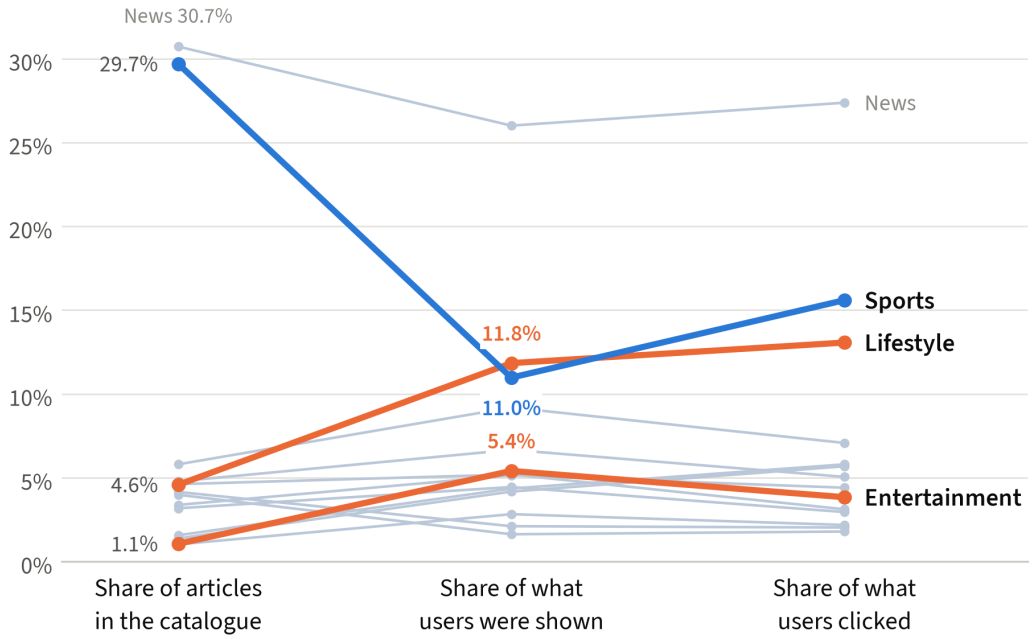
30%
of articles were sports…

11%
…but only this share of what users saw

5.1×
Entertainment: exposure relative to its catalogue share

12%
Lifestyle's share of exposure, from 4.6% of articles

Each category's share of the catalogue, of what users were shown, and of what they clicked



Each line is one of the 14 MIND categories with 50+ articles. Orange means shown more than its catalogue share, blue means shown less, and grey lines are the rest.

When sports did appear, people clicked it more readily than any other large category. Its click-through rate (clicks ÷ times shown) hit 5.75%, and yet it got the least exposure relative to the catalogue (0.37×; measured against articles that were actually shown it's 0.39×, second-lowest after weather). Entertainment went the other way, with one of the lowest click-through rates (2.88%) and the most amplified exposure.

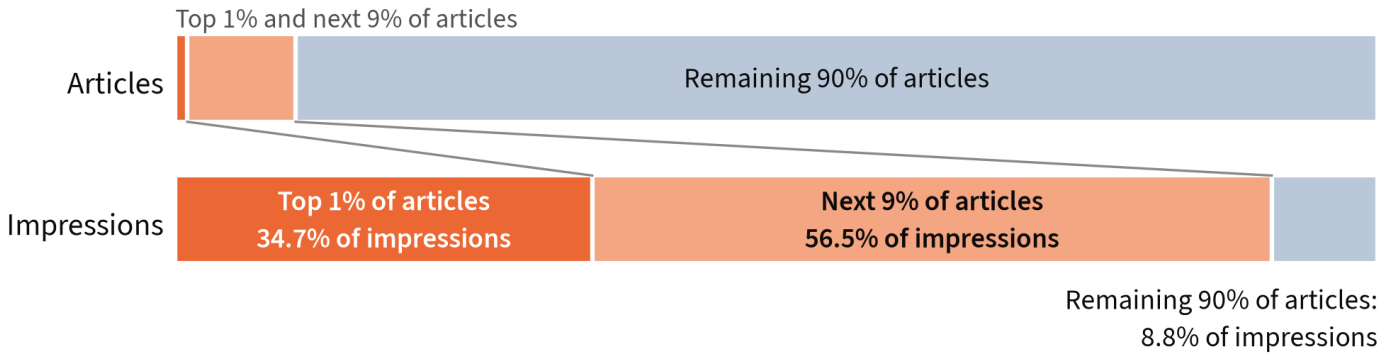
Read this carefully. An *impression* means the log says an article was shown. We use it as a stand-in for exposure, and it doesn't prove anyone read the thing. These numbers come out of the whole Microsoft News system (editors, page layout, recommendation and user navigation together), so they can't tell you anyone intended this, or that it's unfair.

The question we set out to answer: does personalized news exposure cost people diversity? A feed decides which stories millions of people run into every day. If what it shows drifts away from what's available, or keeps narrowing toward whatever each person already clicks, readers end up with a thinner slice of the news and have no way of noticing. Pages 2 and 3 measure that in the real logs; page 4 runs a simulation to see whether a feed can stay relevant while showing more.

WHO GETS THE SPOTLIGHT?

35% of all exposure went to 1% of articles

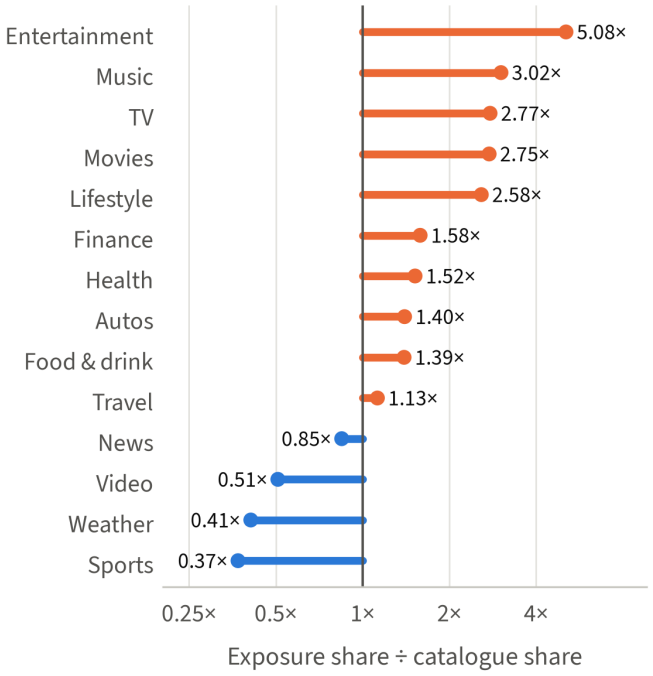
Of the 22,771 articles shown at least once, the 228 most-shown took 34.7% of all impressions and the top tenth took 91.2%, leaving the bottom half with 0.4% between them.

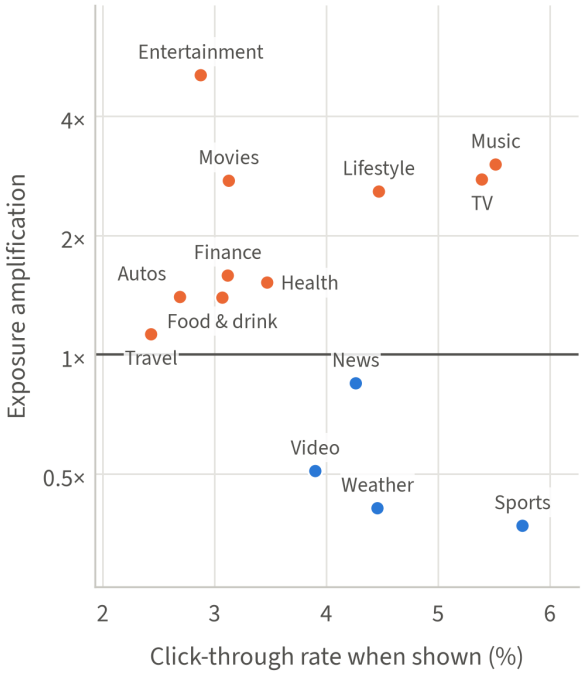


Gini coefficient of impressions per article: 0.92 (0 would mean every article got shown equally, 1 that a single article got everything). Inside any single day it stays between 0.89 and 0.93, so this isn't just old and new articles being lumped together.

Over-exposure has nothing to do with click-through rate

Left: categories ranked by how much exposure they got relative to their catalogue share. Right: that same ratio against how often people clicked each category when it appeared.





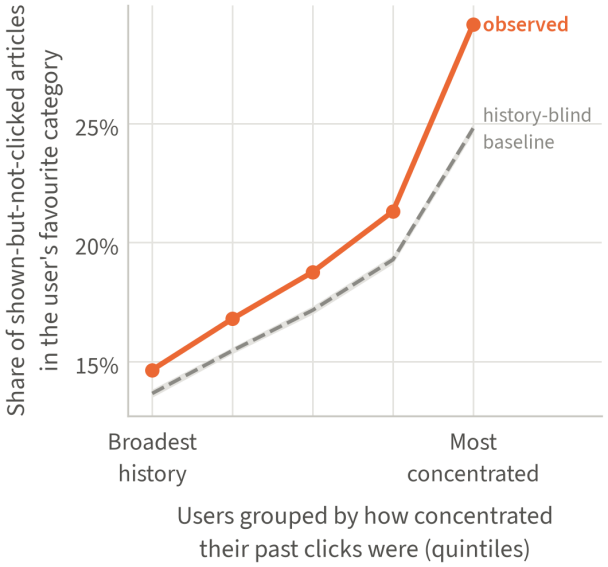
Across the 14 large categories the rank correlation between amplification and click-through rate is ρ = −0.08 (p = 0.78), which is no detectable relationship at all. Click-through rates only compare fairly inside this dataset, because every logged impression contains at least one click and so the logs describe engaged sessions. We left out four categories with fewer than 50 articles.

THE SHRINKING WINDOW?

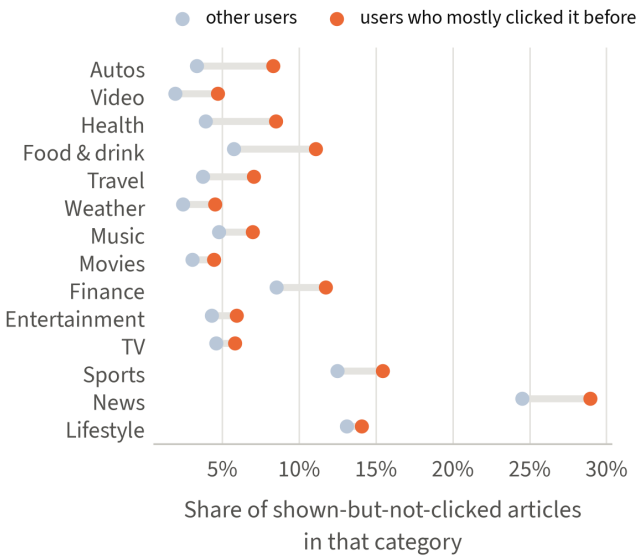
Past favourites show up more than chance predicts

We took 77,216 users with at least 5 clicks *before* the logged week and looked at the articles they were shown but didn't click *during* it. Their most-clicked category made up **20.1%** of those, against roughly 18.1% under a history-blind baseline: a 1.11× tilt (1.15× if you count every article shown). People with the most concentrated click histories saw the biggest tilt (1.18×, against 1.07× for the broadest).

How much of each feed went to the user's favourite

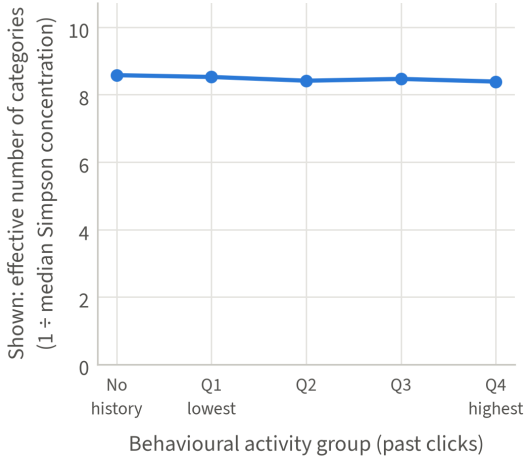


Past fans of a category see more of it than others



Still, the feed stayed wider than the clicks

At the median, a user's past clicks spread across the equivalent of **5.5** evenly used categories, while what they were shown spread across **8.4**. For 75% of users, past clicks were more concentrated than later exposure. Longer click histories went with feeds that were only negligibly more concentrated (rank correlation between history length and exposure concentration, ρ = 0.03).



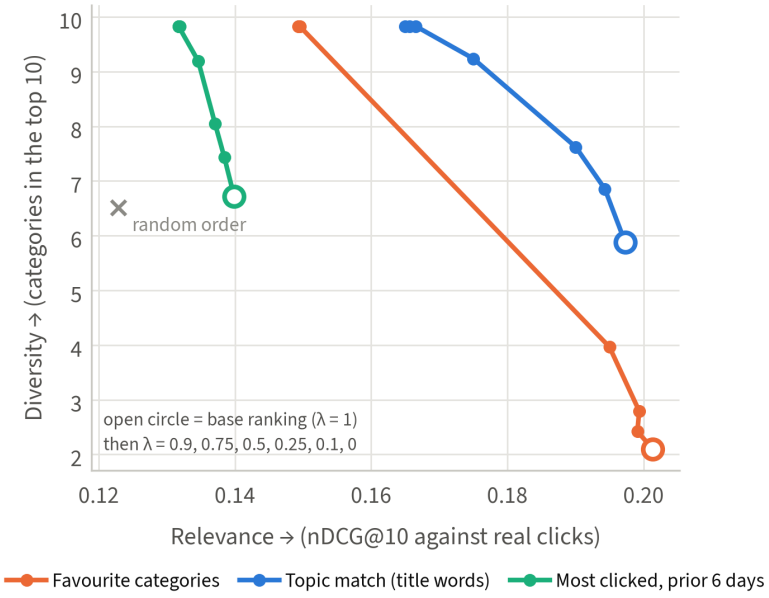
History (clicks from before the logs) comes ahead of the logged exposure (9–15 Nov 2019), but it carries no timestamps, so we can't claim past clicks *caused* what people saw later. Every logged impression contains a click, and clicks lean toward people's interests; that's why we measured the tilt on articles shown but *not* clicked, the cautious option. “Effective categories” is 1 ÷ the median chance that two random items share a category. Groups are based on behaviour only, never demographics.

RELEVANCE VS. DIVERSITY: A SIMULATION

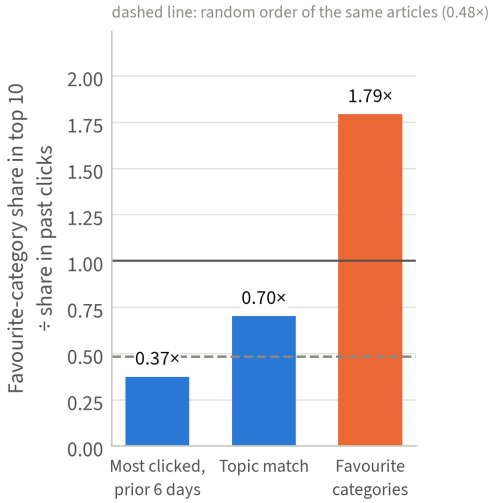
A little diversity costs very little relevance

We built simple recommenders anyone can inspect, then re-ranked each with **Maximal Marginal Relevance** (MMR), which weighs an article's predicted relevance against how similar it is to what's already in the list. Take the most accurate one (favourite-category, narrowly ahead of topic-match on ranking accuracy against real clicks, with nDCG@10 0.201 vs 0.197 and AUC 0.606 vs 0.571): turning the dial to λ = 0.5 moved the number of categories in a top-10 list by **+89%** (2.1 → 4.0) and relevance by **−3.2%**.

Each dot is one setting of the diversity dial



Do recommenders amplify favourites?



The most accurate strategy was also the narrowest. Ranking by past behaviour, the favourite-category strategy fit just 2.1 categories into a top 10 (a random order of the same articles manages 6.5) and stacked the list with each user's favourite at 1.79× its share of their own history. Topic-matching came in at 0.70× and random order at 0.48×; anything above 1× means the list is *narrower* than the person's own clicking.

What we'd do about it. (1) Platforms should publish exposure audits like pages 1 to 3 as a matter of routine: catalogue share against exposure share, how concentrated exposure is, and how far it tilts toward past clicks. (2) The diversity dial was cheap and easy to explain in our simulation, so for Phase 2 we want to prototype a “breadth” control that users adjust themselves, show them plainly what it costs in relevance, and test it with real people.

These are simulated strategies and they don't reproduce MSN's recommender. We tested them on 39,581 real impressions from the last logged day, 15 Nov, each with 20 or more articles shown, re-ranking only what was actually shown and scoring against real clicks (nDCG@10). The models only had information from before that day (pre-period histories plus clicks from the six days prior), and just 34% of that day's candidate articles had any click in those six days.

So does personalized exposure cost diversity?

In the real feed, a modest lean. Among articles people were shown but didn't click, they saw 1.11× as much of their past favourite category as a history-blind baseline predicts, yet the feed stayed wider than their clicks (8.4 vs 5.5 effective categories).

In our simulation, the default narrows. The most accurate strategy fit only 2.1 categories into a top 10.

Adding breadth is cheap. A diversity dial raised categories per top 10 by 89% at a 3.2% relevance cost, in simulation.

METHODS & LIMITATIONS

How we know, and what we can't conclude

Data. Microsoft News Dataset (MIND-small) from Microsoft Research: 65,238 news articles with categories, plus anonymized logs from 94,057 users. That's 230,117 impression logs (9–15 Nov 2019) holding 8,584,442 article impressions and 347,727 clicks.

Definitions. Exposure is an article logged as shown in an impression, a stand-in that doesn't prove anyone read it. **Engagement** is a click. **Catalogue share** = a category's articles ÷ all articles; **exposure share** = its impressions ÷ all impressions; **CTR** = clicks ÷ impressions within that category.

Amplification. Exposure share ÷ catalogue share, where 1× means proportional. 65% of catalogue articles only appear in users' older click histories, so we reran it against just the articles shown at least once, and the category ranking barely moved (rank correlation 0.97).

Concentration. Top-1% and top-10% shares plus the Gini coefficient of impressions per article, recalculated for each day as well.

Narrowing. We used an unbiased Simpson index (the chance two items share a category) alongside entropy, then ran a permutation test on articles shown but not clicked, since every logged impression contains a click and clicks favour people's interests. Shuffling users' favourite categories within groups of similar activity and time period gives the history-blind baseline. The tilt beat that baseline in 8 of 8 alternative setups (history thresholds of 2 to 20 clicks, exposure from 9–14 Nov only or 15 Nov only, and groups of similar activity).

Recommenders. Most clicked over the six days before the test day, TF-IDF similarity between titles and the user's history, and favourite-category preference, each re-ranked with MMR as λ went from 1 down to 0. We scored them on 15 Nov using nDCG@10 and AUC (ranking accuracy against real clicks), distinct categories, intra-list diversity, coverage and an amplification ratio. No 15 Nov click ever went into a model.

Limitations

- Observational data, so these are associations rather than causes.
- One week of US Microsoft News. It says nothing direct about the wider internet, other platforms, Canada or other years.
- Users are anonymized. No demographic or location data exist and we didn't infer any; our groups are purely behavioural.
- An impression shows an article was logged as displayed, not that anyone saw or read it.
- Every logged impression contains a click, which makes these engaged sessions; click-through rates compare fairly only inside this dataset.
- Click histories come from before the logs, but they carry no timestamps and don't update during the logged week.
- Observed exposure is the output of MSN's whole operation (editors, layout, recommendation, navigation), not any single algorithm.
- Our recommenders are simulations run on logged candidate lists and don't reproduce MSN's own system.
- We measured diversity with 18 broad categories, which misses viewpoint and source diversity. Title-word diversity (ILD) sat near its maximum for every strategy, so we didn't draw conclusions from it.
- MIND-small samples 50,000 users per split; it isn't a census of MSN traffic.

Sources. Wu, F. et al. (2020). MIND: A Large-scale Dataset for News Recommendation. *Proceedings of ACL 2020*, 3597–3606. Dataset: Microsoft News Dataset (MIND), Microsoft Research, https://msnews.github.io/ (Microsoft Research License Terms). Carbonell, J. & Goldstein, J. (1998). The use of MMR, diversity-based reranking for reordering documents and producing summaries. *SIGIR '98*. **Reproducibility.** Every number traces back to code, notebooks and result files at https://github.com/suryalionael/information-diversity. We don't redistribute the dataset itself.